

PoMAS

Model Implementation and Experimental Findings Report

Offline EEG seizure detection and prediction research prototype

Evidence cut-off: 16 August 2026

Primary dataset: CHB-MIT Scalp EEG

Evaluation scope: retrospective, offline, non-clinical

Evidence policy

Metrics in this report are read from saved JSON artifacts. Console-only trial values that were overwritten are not treated as final evidence. Corrected global-v2 experiments are separated from older exploratory per-file results.

Prepared as a technical record of implementation progress, ablation findings, limitations, and the next defensible research steps.

1 Executive summary

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Saved experiment results reviewed

241,552

Clean global-v2 windows across 8 subjects

0.670 +/- 0.245

EfficientNet 5-fold mean ROC-AUC

PoMAS now has a reproducible preprocessing foundation and implemented models for detection, patient-dependent prediction, and patient-independent prediction. The engineering pipeline works end to end, including CUDA training, protected checkpoints, validation-only threshold selection, and event consolidation. The central scientific result is more cautious: model performance changes sharply across recordings and subjects, so generalization has not been established.

- The leakage-safer patient-dependent EfficientNet-SVM experiment achieved mean test ROC-AUC 0.670 +/- 0.245 across five seizure-cluster folds, but fold ROC-AUC ranged from 0.343 to 0.985.
- The best patient-independent result on chb05 was the dual-graph GNN (ROC-AUC 0.637, PR-AUC 0.097), but the reciprocal held-out chb03 result fell to ROC-AUC 0.408 and PR-AUC 0.068.
- MFCC-Siamese, seizure-only MFCC, SALT-inspired detection, and LaBraM transfer did not provide robust preictal generalization under the tested splits.
- Strong ranking metrics from early single-split experiments were often paired with zero sensitivity or failed validation, demonstrating why validation/test separation and repeated folds are essential.
- The current system must be described as an offline research prototype. It is not clinically validated and is not suitable for treatment, EHR, or neurostimulation decisions.

Overall conclusion

Implementation is technically functional and proposal-aligned at prototype level. Patient-dependent prediction shows limited but unstable signal. Patient-independent prediction remains subject-dependent and inconclusive. The next implementation priority is the offline agent/event pipeline, followed by broader subject-level evaluation rather than additional tuning on chb03 or chb05.

2 Data foundation and evaluation protocol

The corrected global-v2 archives label each subject on a continuous summary-clock timeline, allowing a 30-minute preictal interval to cross EDF recording boundaries. Windows are 10 seconds long with a 5-second stride, 19 bipolar channels, and 256 Hz sampling. Ictal windows are excluded from early-warning prediction; the 30-minute postictal recovery interval is excluded from ordinary interictal training.

Subject	Windows	Interictal	Preictal	Ictal	Archive status
chb01	27,118	24,558	2,460	100	Complete
chb02	24,494	23,377	1,077	40	Complete
chb03	25,054	22,750	2,211	93	Complete
chb05	26,250	24,333	1,796	121	Complete
chb06	44,777	41,151	3,577	49	Complete
chb07	47,429	46,278	1,080	71	Complete
chb08	12,846	10,852	1,800	194	Complete
chb10	33,584	30,828	2,653	103	Complete
chb12	12,156	7,395	4,517	244	Excluded: incomplete montage

chb12 exclusion

chb12_27.edf, chb12_28.edf, and chb12_29.edf did not support the required montage. Those files contain 13 seizures, so chb12 was preserved as an incomplete archive but excluded from the main six-subject training cohort.

Primary patient-independent split

Role	Subjects	Purpose
Training	chb01, chb02, chb06, chb07, chb08, chb10	Balanced sampling across subjects and classes
Validation	chb03	Checkpoint and threshold selection only
Test	chb05	Unseen-subject evaluation
Reciprocal	Validation chb05; test chb03	Checks whether performance is subject-specific

Metric interpretation

- ROC-AUC evaluates ranking over thresholds; PR-AUC is more informative under preictal class imbalance.
- Sensitivity and specificity depend on a threshold selected exclusively on validation data.
- False-positive windows/hour counts overlapping windows and is not a clinical alarm rate.
- Alarm events are consolidated with a 30-minute refractory period. Preictal-episode sensitivity is not labelled seizure-level sensitivity because adjacent seizures can share one preictal episode.

3 Implemented model inventory

Model	Task	Regime	Data	Status
Temporal LSTM	Prediction	Patient-dependent	Legacy per-file	Completed baseline
Five-band spectral logistic model	Prediction	Patient-dependent	Legacy per-file	Completed baseline
Multisubject spectral logistic model	Prediction	Patient-independent	Legacy per-file	Completed baseline
LTformer-inspired detector	Detection	Patient-independent	Legacy per-file	Completed; not exact SALT
LTformer-inspired patient model	Detection	Patient-dependent	Legacy per-file	Completed; not exact SALT
LaBraM transfer detector	Detection	Patient-independent	LaBraM 200 Hz	Completed transfer trial
EfficientNet-B0 + SVM	Prediction	Patient-dependent	Legacy per-file	1- and 5-epoch trials
EfficientNet seizure-cluster CV	Prediction	Patient-dependent	Global-v2 chb01	Five folds complete
MFCC-Siamese	Prediction	Patient-independent	Global-v2	2-subject and 6-subject
MFCC seizure-only CNN	Prediction	Patient-independent	Global-v2	Auxiliary-loss ablation
Dual-graph GNN	Prediction	Patient-independent	Global-v2	Forward and reciprocal tests

Naming discipline

The custom LSTM-Transformer is reported as LTformer-inspired, not as a faithful SALT reproduction. The MFCC and GNN models are documented as 19-channel PoMAS adaptations because the cited papers used different channel representations and protocols.

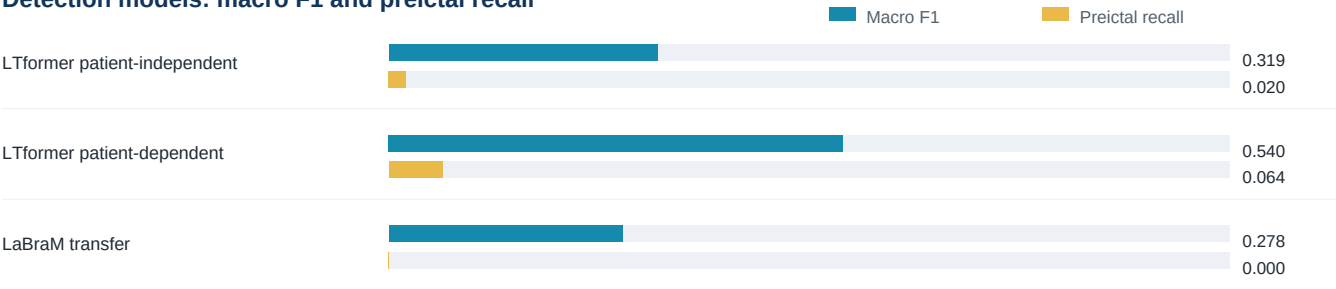
What has not yet been completed

- A production-quality offline agent event store and replay orchestrator.
- Validated SHAP, Grad-CAM, or attention explanations connected to final predictions.
- A dashboard driven by real replay outputs rather than demonstration values.
- Siena external-dataset preprocessing and evaluation.
- Exact SALT self-supervised pretraining, exact six-SVM EfficientNet reproduction, or exact cited GNN/MFCC protocols.

4 Detection model results

Model	Accuracy	Macro F1	Preictal recall	Evaluation note
LTformer-inspired, patient-independent	0.788	0.319	0.020	chb01/02 train; chb03 val; chb05 test
LTformer-inspired, patient-dependent	0.457	0.540	0.064	Recording split within chb01
LaBraM transfer detector	0.693	0.278	0.000	219 pretrained tensors loaded

Detection models: macro F1 and preictal recall



Detection finding Accuracy overstated utility because interictal windows dominate. The patient-independent LTformer reached accuracy 0.788 but preictal recall only 0.020. LaBraM loaded 219 pretrained tensors yet produced zero preictal recall. The patient-dependent LTformer improved macro F1 to 0.540 but preictal recall remained 0.064.

An earlier one-epoch LaBraM console diagnostic was observed but was not retained as a separate result artifact. The saved final LaBraM JSON is therefore the only LaBraM value used in the comparison table.

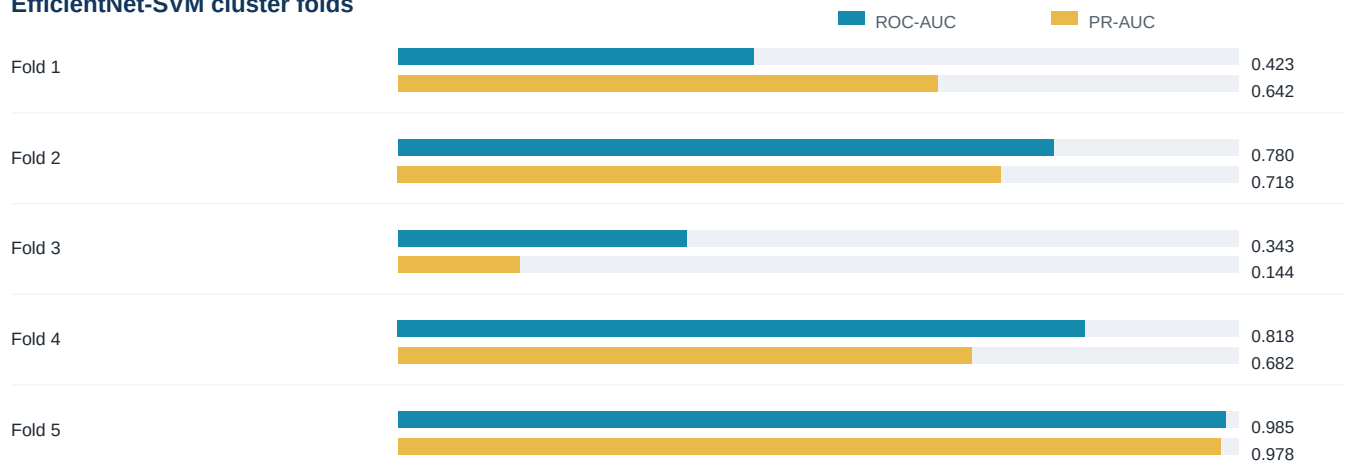
5 Patient-dependent prediction results

Model	ROC-AUC	PR-AUC	Sensitivity	Specificity	Interpretation
Temporal LSTM	0.771	0.798	0.000	1.000	Threshold selected 0.95; all-negative test decision
Spectral logistic	0.899	0.921	0.000	1.000	High test ranking; validation ROC-AUC 0.107
EfficientNet-SVM, 1 epoch	0.858	0.723	0.082	0.980	Validation ROC-AUC 0.300
EfficientNet-SVM, 5 epochs	0.849	0.711	0.088	0.976	Validation ROC-AUC 0.295; no gain from epochs
EfficientNet-SVM, 5-fold cluster CV	0.670 +/- 0.245	0.633 +/- 0.271	0.661	0.635	Primary leakage-safer patient-dependent estimate

Seizure-cluster cross-validation

Fold	Test cluster	ROC-AUC	PR-AUC	Sensitivity	Specificity	FP windows/h
Fold 1	1	0.423	0.642	0.245	0.631	265.8
Fold 2	2	0.780	0.718	0.566	0.790	151.3
Fold 3	3	0.343	0.144	0.703	0.244	544.4
Fold 4	4	0.818	0.682	0.897	0.561	316.3
Fold 5	5	0.985	0.978	0.894	0.948	37.6

EfficientNet-SVM cluster folds



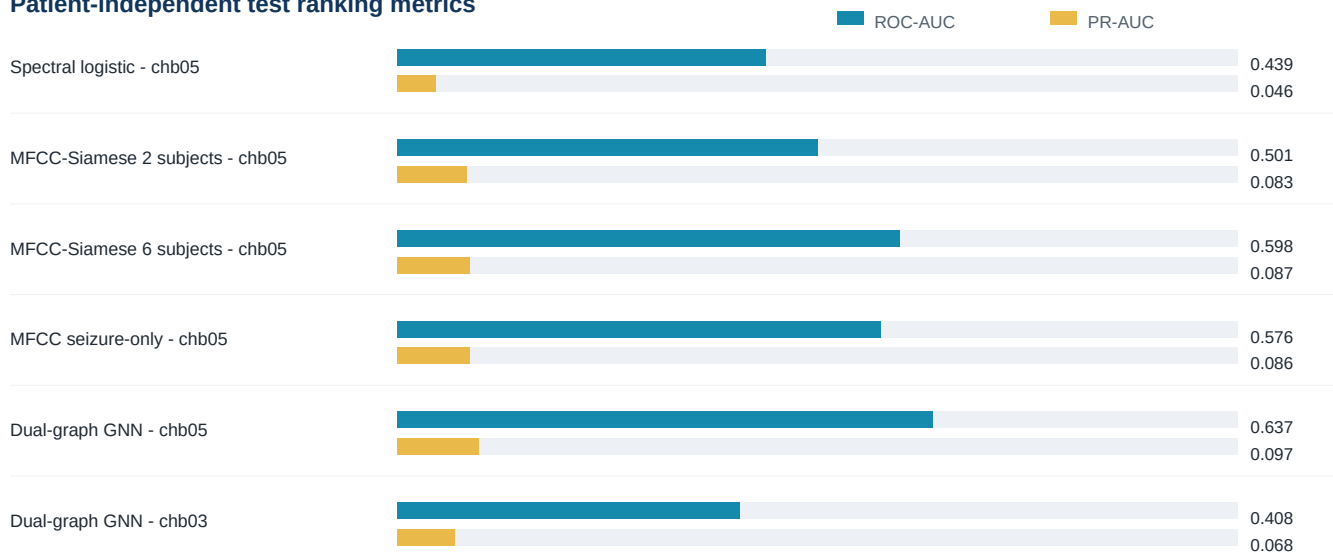
Patient-dependent finding

Single-split test AUC values above 0.84 were not supported by validation. Five-fold cluster evaluation is more defensible and shows substantial nonstationarity: ROC-AUC ranges from 0.343 to 0.985 and mean false-positive windows/hour is 263 +/- 170.

6 Patient-independent prediction results

Model	Test	ROC-AUC	PR-AUC	Balanced acc.	Episode sens.	False alarms/h
Multisubject spectral logistic	chb05	0.439	0.046	-	-	-
MFCC-Siamese, 2 train subjects	chb05	0.501	0.083	0.500	0.000	0.000
MFCC-Siamese, 6 train subjects	chb05	0.598	0.087	0.534	0.714	2.101
MFCC seizure-only ablation	chb05	0.576	0.086	0.542	0.714	1.953
Dual-graph GNN, forward	chb05	0.637	0.097	0.568	0.714	1.775
Dual-graph GNN, reciprocal	chb03	0.408	0.068	0.477	0.778	2.089

Patient-independent test ranking metrics



Patient-independent finding

The GNN is the best available branch on chb05 (ROC-AUC 0.637), but the reciprocal held-out chb03 test falls below chance (ROC-AUC 0.408). This proves subject-specific performance and prevents a claim of stable patient-independent prediction.

7 Ablation study and causal findings

Ablation A: training-subject diversity

MFCC configuration	Train subjects	Test ROC-AUC	Test PR-AUC	Outcome
Siamese	2	0.501	0.083	Chance-level test ranking and zero sensitivity at validation threshold
Siamese	6	0.598	0.087	ROC improved, but specificity collapsed to 0.098

Adding four clean training subjects increased MFCC-Siamese test ROC-AUC from 0.501 to 0.598, but it did not create a useful operating point. The model produced 71 false alarm events on chb05 and 2.10 false alarms/hour.

Ablation B: Siamese auxiliary objectives

Configuration	Patient loss	Contrastive loss	Val ROC	Test ROC	Test PR	False alarms/h
MFCC-Siamese	0.2	0.1	0.439	0.598	0.087	2.101
Seizure-only MFCC	0	0	0.456	0.576	0.086	1.953

Removing patient-identity and contrastive losses did not materially improve PR-AUC or false alarms. The primary limitation is therefore the cross-subject MFCC representation and domain shift, not just the Siamese loss formulation.

Ablation C: reciprocal GNN holdout

Validation	Test	Test ROC-AUC	Test PR-AUC	Balanced acc.	False alarms/h
chb03	chb05	0.637	0.097	0.568	1.775
chb05	chb03	0.408	0.068	0.477	2.089

Causal conclusion

Performance follows the held-out subject more than the model-selection role: chb05 is consistently easier, while chb03 remains difficult. The dominant problem is inter-subject distribution shift.

8 Cross-experiment findings and limitations

Findings supported by the completed experiments

- Correct global labels and subject/recording separation materially change the credibility of reported performance. Early single-split scores cannot be treated as final estimates.
- More training epochs are not automatically beneficial. EfficientNet test ROC-AUC decreased from 0.858 at one epoch to 0.849 at five epochs while validation remained poor.
- More training subjects help some ranking metrics but do not remove subject-domain shift. MFCC and GNN results remain unstable across unseen patients.
- Thresholded sensitivity can be misleading. MFCC-Siamese reached test sensitivity 0.970 only by reducing specificity to 0.098 and generating 71 false alarm events.
- A zero false-alarm result may represent an all-negative classifier. The two-subject MFCC model had zero test alarms because its validation threshold exceeded every test score.
- AUC and PR-AUC must be considered with validation behavior, class prevalence, sensitivity, specificity, and event-level false alarms.

Current limitations

Area	Limitation
Cohort	Eight clean global-v2 subjects are available; this remains a small subset of CHB-MIT.
External validity	Siena has not yet been implemented or evaluated.
Protocol fidelity	SALT, EfficientNet-SVM, MFCC, and GNN are proposal-aligned adaptations, not exact paper reproductions.
Alarm definition	The 30-minute refractory event metric is an offline prototype, not a clinically validated SOP/SPH protocol.
Model selection	chb03 and chb05 have now both been used as validation/test subjects; further tuning against them risks overfitting.
Explainability	Existing XAI modules are not yet validated against the final model outputs.
Deployment	No clinical, EHR, wearable, or stimulation integration has been validated.

Interpretation boundary

The project demonstrates implementation feasibility and exposes generalization failure under stricter evaluation. It does not demonstrate reliable seizure forecasting, clinical efficacy, or readiness for deployment.

9 Proposal alignment and implementation status

Component	Proposed method	Status	Evidence/qualification
Preprocessing	19 channels, filtering, normalization, 10-second windows	Implemented	Global-v2 labels span EDF boundaries
Detection	SALT-style LSTM-Transformer	Partially implemented	LTformer-inspired; no full dual-stream SSL
Patient-dependent prediction	STFT + correlation + EfficientNet-B0 + SVM	Implemented adaptation	Single SVM and 19 channels; five-fold cluster CV complete
Patient-independent MFCC	MFCC-CNN/Siamese	Implemented adaptation	Temporal MFCC maps, 2- and 6-subject studies, loss ablation
Patient-independent GNN	Physical and subject-specific graph information	Implemented adaptation	Shared-electrode physical graph + functional correlation graph
Multi-agent/CEP	Monitoring, prediction, alert agents	Scaffold only	Offline persistent replay is next
XAI	SHAP, Grad-CAM, attention	Scaffold only	Not yet validated with final prediction path
Dashboard	Risk timeline and tiered alerts	Demonstration only	Current dashboard is not connected to real replay
External validation	CHB-MIT and Siena	CHB-MIT only	Siena remains optional stretch work

Scope decision

PoMAS should remain an offline, retrospective research prototype. Simulated alerts, event logs, and decision-support demonstrations are in scope. Live EHR integration, clinician notification, neurostimulation, and medical-device claims are out of scope.

What is defensible to claim now

- A modular PoMAS research pipeline has been implemented from raw CHB-MIT EDF files to protected model artifacts and repeatable evaluation.
- Patient-dependent and patient-independent modelling branches have been compared under increasingly strict data splits.
- Corrected evaluation reveals substantial recording and subject nonstationarity that earlier single-split scores obscured.
- Negative and unstable results are retained as formal ablations rather than discarded.

10 Conclusions and recommended next work

Final conclusions

#	Conclusion	Evidence
1	The data pipeline is the strongest completed contribution.	Global timeline labels, protected HDF5 archives, and disjoint splits provide a reproducible experimental base.
2	Patient-dependent prediction contains signal but is unstable.	EfficientNet-SVM five-fold ROC-AUC averages 0.670 with very high fold variance and false-positive windows.
3	Patient-independent prediction is not established.	GNN and MFCC results change materially between chb03 and chb05. Reciprocal GNN testing demonstrates subject-domain shift.
4	Accuracy alone is inadequate.	Macro F1, preictal recall, PR-AUC, validation behavior, and false alarm events expose failures hidden by accuracy or isolated AUC values.
5	Further tuning on the same held-out subjects should stop.	chb03 and chb05 have served as both validation and test targets. Additional tuning would weaken the evidential boundary.
6	The remaining thesis value is integration and rigorous reporting.	Offline agents, persistent events, reproducible replay, and transparent limitations are higher priority than another model search.

Recommended implementation order

Priority	Work item	Acceptance criterion
1	Offline agent replay and persistent event store	Replays saved predictions in timestamp order; logs monitoring, risk, and simulated alerts reproducibly
2	CEP alarm rules	Consecutive-window logic, refractory period, thresholds, and alert tiers are tested and auditable
3	Unified comparison artifact	One machine-readable table covering all model splits, metrics, thresholds, and caveats
4	Validated explanation for selected branch	Explanation corresponds to the actual final score path and passes sanity checks
5	Dashboard connection	Displays real offline replay events; no random risk values
6	Optional expanded subject CV or Siena	Attempt only after core integration and thesis chapters are secure

Recommended thesis framing

PoMAS is a proof-of-concept multi-agent framework evaluated retrospectively on public EEG. Its contribution is a reproducible integration and evaluation workflow that identifies when apparently strong window-level models fail under recording-level, seizure-cluster, and subject-level generalization tests.

Appendix A Saved evidence artifacts

Experiment	Saved metric artifact
Temporal LSTM	output/baseline_chb01_results.json
Spectral patient-dependent	output/spectral_baseline_results.json
Spectral patient-independent	output/multisubject_spectral_test_chb05.json
LTformer patient-independent	output/salt_ltformer_detector_results.json
LTformer patient-dependent	output/salt_ltformer_chb01_patient_dependent_results.json
LaBraM transfer	output/labram_detector_results.json
EfficientNet 1 epoch	output/efficientnet_chb01_prediction_results_1epoch.json
EfficientNet 5 epochs	output/efficientnet_chb01_prediction_results.json
EfficientNet cluster CV	output/efficientnet_svm_cluster_cv_chb01/cross_validation_results_all.json
MFCC-Siamese 2 subjects	output/mfcc_siamese_global_v2/results.json
MFCC-Siamese 6 subjects	output/mfcc_siamese_6train_global_v2/results.json
MFCC seizure-only	output/mfcc_cnn_6train_global_v2/results.json
GNN forward	output/gnn_6train_global_v2/results.json
GNN reciprocal	output/gnn_6train_reverse_global_v2/results.json

Primary methodological references

- CHB-MIT Scalp EEG Database. PhysioNet: <https://physionet.org/content/chbmit/1.0.0/>
- Saadoon et al. EfficientNet-B0 and SVM seizure prediction. DOI: 10.3390/bioengineering12020109
- Dissanayake et al. Patient-independent seizure prediction using scalp EEG. DOI: 10.1109/JSEN.2021.3057076
- Dissanayake et al. Geometric deep learning for subject-independent prediction. DOI: 10.1109/JBHI.2021.3100297
- Xiao et al. SALT seizure detection. DOI: 10.1142/S0129065726500127
- LaBraM official implementation: <https://github.com/935963004/LaBraM>

Report generation note: all quantitative values were loaded from the saved artifacts listed above. Values are rounded for presentation; source JSON files retain full precision.